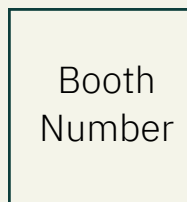
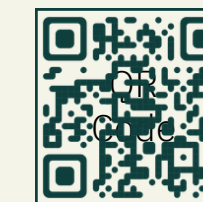




SADIN: An Arabic-First, On-Device Protective Middleware That Stops Sensitive Enterprise Data Leaking into Generative AI Tools

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Introduction

Generative AI is now part of daily work. An employee stuck on a problem pastes a device configuration, uploads an internal document or screenshots an admin console into an AI assistant. The employee means no harm, but the data now sits with the AI vendor.

From there attackers can reach it without breaking into the organisation: through stolen AI accounts that keep the full chat history (Group-IB found ChatGPT logins in more than 225,000 infostealer logs in 2023), through chats shared by link that ended up in Google results (2025), and through data the vendor retains (Samsung, 2023).

Blocking AI sites does not solve this. Usage moves to personal devices and accounts, and the organisation loses both visibility and the productivity gains. SADIN is a protective middleware: everything an employee sends to an AI tool passes through it, on the device or on the company's own servers. Employees keep using AI, and the company's secrets stay inside.

Objectives:

- (1) intercept text, files and images before they leave;
- (2) detect what must never leave: classified documents and Arabic classification stamps, credentials, internal IPs, hostnames and ports, and account identifiers;
- (3) redact instead of block;
- (4) tie AI use to the work identity and log every decision.

Methodology

Design approach: one policy engine, three enforcement points, inspection on the device or on-premises only.

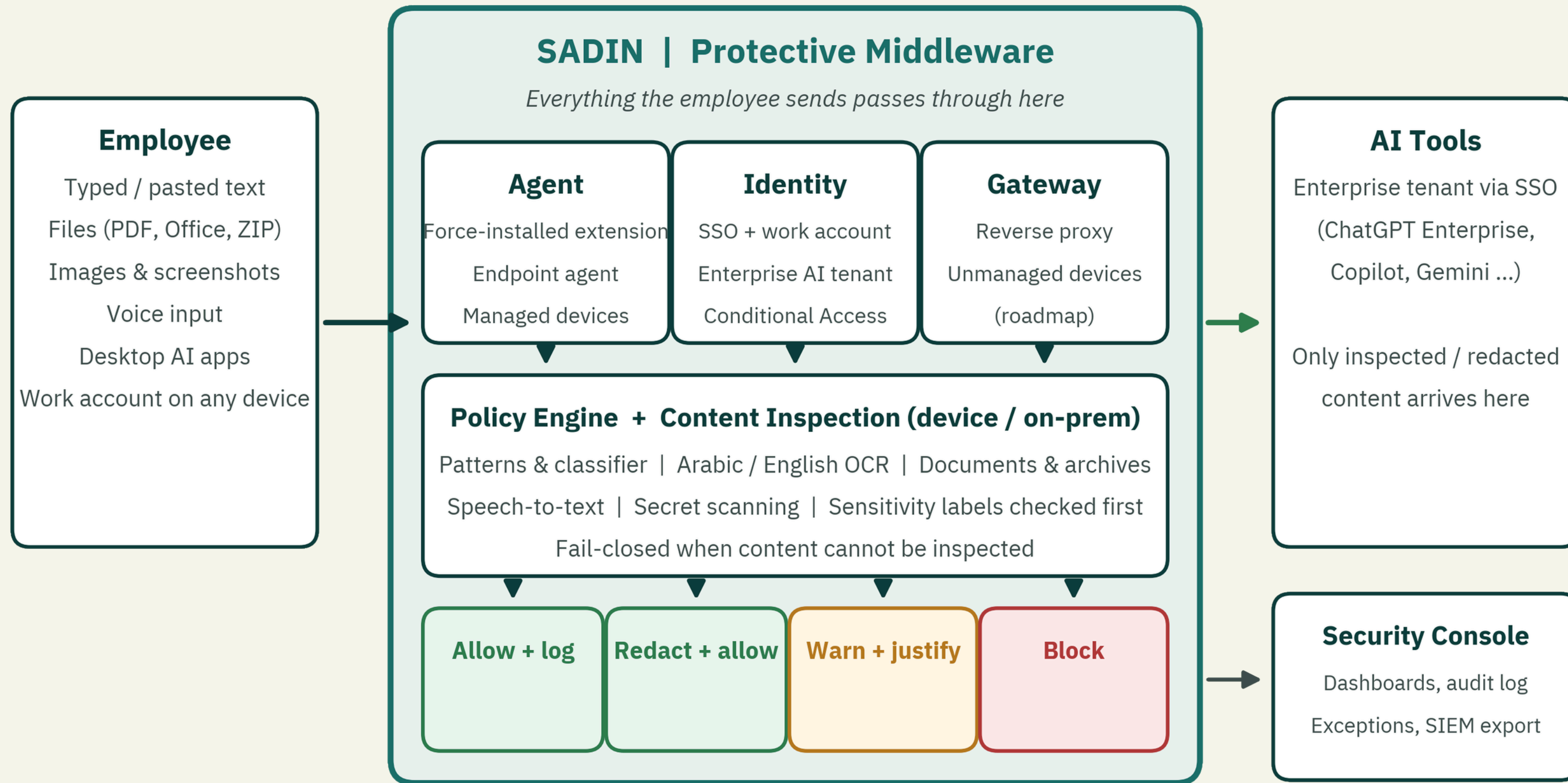
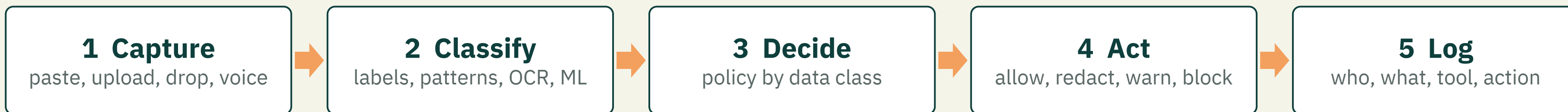


Figure 1. SADIN reference architecture: enforcement points, policy engine and the four policy actions.



Built and tested: Chrome/Edge extension (Manifest V3) | regex, range and entropy validators, Arabic and English patterns | on-prem service with Arabic/English OCR | decision log, CEF export to SIEM
Next: endpoint agent | ML classifier | document fingerprinting | enterprise AI tenant | gateway mode

Results

Table 1. Channels and prototype status

Channel	Prototype status
Typed or pasted text	Working
File upload, drag and drop	Working (text files)
Images and screenshots	Working (on-prem OCR)
Office files, PDF, archives	Blocked, fail-closed
Desktop AI apps, IDE assistants	Design: endpoint agent
Voice input	Design: speech-to-text
Account mismatch	Design: Identity + Gateway

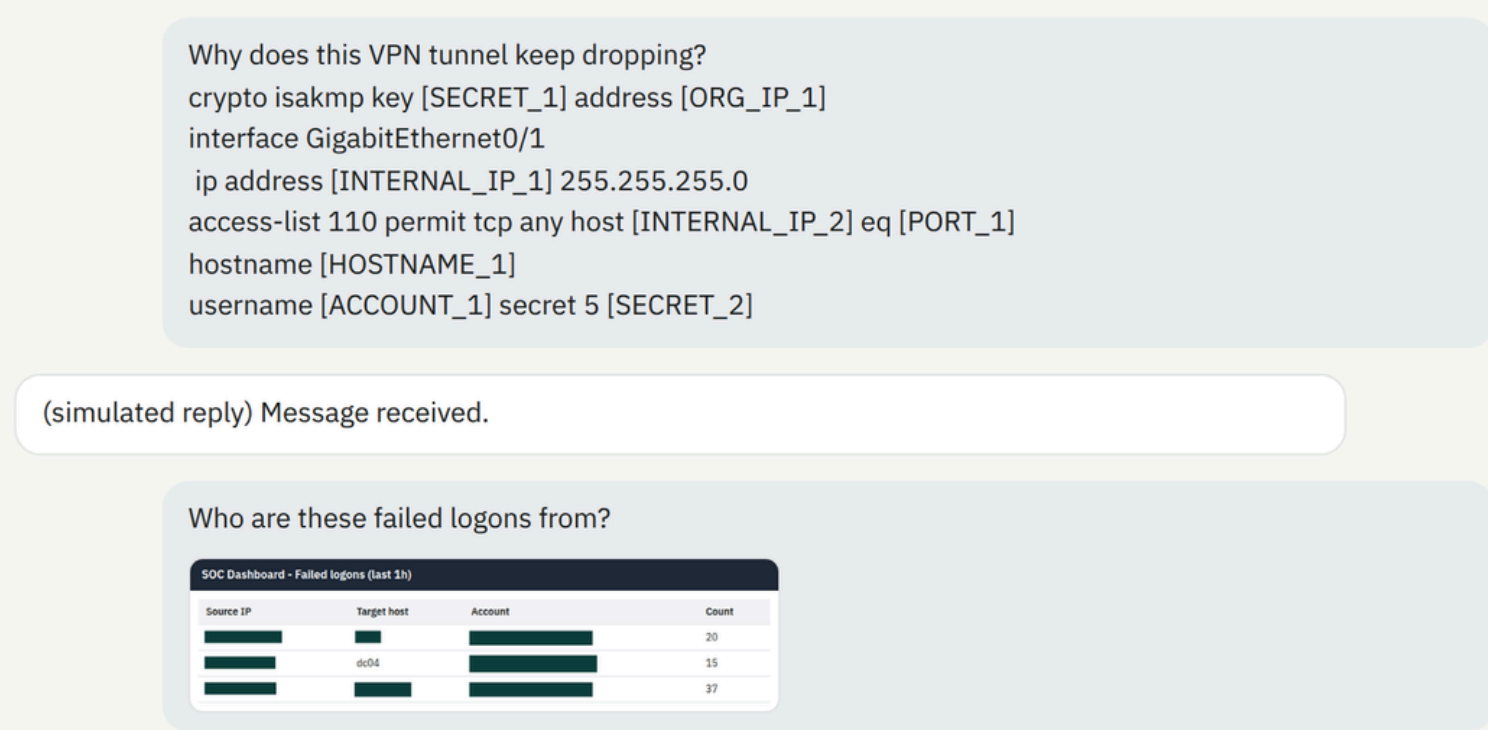


Figure 2. What the AI tool receives after SADIN. One hostname (dc04) was missed; see Table 3.

Table 3. Measured results (unseen test sets, first run)

Metric	Measured
Sensitive items caught in text	86.3% (44 of 51)
False positives on clean prompts	0 of 50
Screenshot values masked; classified Arabic memos blocked	95.8% (69 of 72); 8 of 8 (4 stamped)
Time: message / 1 MB paste / screenshot	<0.1 ms / 0.2 s / 1.7 s
Arabic items caught (Arabic held-out set)	100% (36 of 36); 1 false positive in 40

Table 2. Scenarios demonstrated in the prototype

Employee action	SADIN outcome (verified)
Pastes firewall config to troubleshoot	IPs, hostname, port, key redacted
Asks in Arabic with an IP, port, user and password	All four redacted; question intact
Uploads a .env config file	Secrets removed from the copy
Pastes SOC dashboard screenshot	OCR finds data, regions masked
Pastes memo with Arabic Secret stamp	Blocked, reason shown

Innovation

SADIN is a protective middleware that combines four properties we did not find together in the tools we reviewed: (1) it reads Arabic, including classification stamps on scanned pages; (2) it inspects on the employee's device or the company's own server; (3) it redacts the sensitive part instead of blocking the whole task; (4) its policies map to NCA ECC-2:2024 and the NDMO levels.

Table 4. Comparison with existing approaches

Capability	Cloud DLP / CASB	SADIN
Arabic OCR, classification stamps	Not found in the tools we reviewed	Built in and tested
Where inspection runs	Vendor cloud	Device or on-prem
Images and files	Limited	Images masked; files in Phase 2
Desktop AI apps and voice	Browser only	Endpoint agent (Phase 2)
Sanctioned AI via work email	Separate project	Part of the design
NCA ECC / NDMO mapping	Generic	Mapped by design

86.3%

of sensitive items caught (held-out set, first run)

0

false positives in 50 clean prompts

95.8%

of screenshot values masked

Conclusion

Gap closed

AI use moves from an invisible risk to an authorised, inspected and logged activity, without taking the tools away from employees.

Impact

Organisations adopt AI without leaving classified documents, credentials or network details with an AI vendor, where a stolen account or a shared link could expose them.

Readiness

Working prototype: Chrome/Edge extension plus on-prem OCR service, tested end to end on text, files and screenshots, with the test sets and raw results in the project folder. Endpoint agent and enterprise tenant next.

SADIN keeps AI available to employees, makes every request authorised, inspected and logged, and keeps the sensitive data inside the company.

Future Work & References

- Endpoint agent with Office/PDF parsing and MIP sensitivity-label integration.
- Enterprise AI tenant APIs and Conditional Access playbook; personal-account blocking.
- Gateway mode (reverse proxy) for unmanaged devices, then a pilot with a Saudi organisation.

References: NCA, ECC-2:2024 | NDMO, Data Classification Policy | Group-IB, stolen ChatGPT accounts (2023) | Samsung ChatGPT incident (2023) | Shared ChatGPT chats in Google (2025) | OWASP Top 10 for LLM Applications. The QR code in header contains the full reference list.